

CSE331 Theory Assignment 01

1.

$L_1 = \{ w \in \{0, 1\}^* \mid w \text{ does not contain "10110" in the substring} \}$

$L_2 = \{ w \in \{0, 1\}^* \mid w \text{ ends with "10110"} \}$

$L_3 = \{ w \in \{0, 1\}^* \mid w \text{ accepts at least three "00", considering no overlapping} \}$

$L_4 = \{ w \in \{0, 1\}^* \mid w \text{ accepts at most three "00", considering no overlapping} \}$

$L_5 = \{ w \in \{0, 1\}^* \mid w \text{ starts and ends with different symbol} \}$

$L_6 = \{ w \in \{0, 1\}^* \mid w \text{ contains even number of "0"s and odd number of "1"s} \}$

$L_7 = \{ w \in \{0, 1, 2\}^* \mid w = 0^m 1^n 2^k \text{ where } m, k \geq 2 \text{ and } n \geq 1 \}$

$L_8 = \{ w \in \{0, 1\}^* \mid w = 0^m 1^n \text{ where } m \text{ is odd and } n \geq 1 \}$

$L_9 = \{ w \in \{0, 1\}^* \mid \text{contains } 01^m 0 \text{ as a substring where } m \text{ is divisible by } 3, m \geq 2 \}$

$L_{10} = \{ w \in \{0, 1\}^* \mid |w| = 4n + 2, \text{ where } n \geq 1 \}$

$L_{11} = \{ w \in \{0, 1\}^* \mid w \text{ equal number of occurrences of substring "01" and "10"} \}$

$L_{12} = \{ w \in \{0, 1\}^* \mid w \text{ is any string not in } 0^* 1^* \}$

$L_{13} = \{ w \in \{0, 1\}^* \mid \text{every even position letter in } w \text{ is different from first letter of } w \}$

$L_{14} = \{ w \in \{0, 1\}^* \mid \text{the decimal equivalent of } w \text{ is a multiple of } 5 \}$

$L_{15} = \{ w \in \{0, 1\}^* \mid \text{every 0 is followed by at least three '1's} \}$

$L_{16} = \{ w \in \{a, b\}^* \mid w \text{ has "aab" somewhere in the preceding and starts with "ba"} \}$

$L_{17} = \{ w \in \{0, 1\}^* \mid \text{after the first 1 in } w, \text{ the number of 0's is even} \}$

$L_{18} = \{ w \in \{0, 1\}^* \mid \text{if } w \text{ starts with 0, it contains exactly three "1"s} \}$

$L_{19} = \{ w \in \{0, 1\}^* \mid \text{every 5th symbol from LHS is 0} \}$

$L_{20} = \{ w \in \{0, 1\}^* \mid \text{third symbol from RHS is "1"} \}$

a) Draw the state diagram of DFA for L_1 to L_{19}

2. Draw the regex for the following regular language:

$L_1 = \{ w \in \{0, 1\}^* \mid w \text{ starts with "1" and ends with "01"} \}$

$L_2 = \{ w \in \{0, 1\}^* \mid \text{third last symbol is "1"} \} =$

$L_3 = \{ w \in \{0, 1\}^* \mid \text{2nd last symbol is "0" and 7th last symbol is 1} \}$

$L_4 = \{ w \in \{0, 1\}^* \mid w \text{ accepts at least three "00"} \}$

$L_5 = \{ w \in \{0, 1\}^* \mid w \text{ contains atleast two "01"} \}$

$L_6 = \{ w \in \{0, 1\}^* \mid \text{every 0 is followed by at least three '1's} \}$

$L_7 = \{ w \in \{0, 1\}^* \mid \text{the number of "11" is multiple of 5, considering}$

non-overlapping}

$L_8 = \{ w \in \{0, 1\}^* \mid \text{there is no consecutive '0's} \}$

$L_9 = \{ w \in \{0, 1\}^* \mid w \text{ does not contain "10"} \}$

3. Complete the following questions. This question is a continuation of qs 1, so L_1 refers to the L_1 of qs 1.

$L_{21} = \{ w \in \{0, 1\}^* \mid \text{length is exactly three} \}$

$L_{22} = \{ w \in \{0, 1\}^* \mid \text{every even position is '1'} \}$

$L_{23} = \{ w \in \{0, 1\}^* \mid \text{'10' appears even number of times as a substring} \}$

$L_{24} = \{ \text{'000', '010', '100', '110'} \}$

$L_{25} = \{ w \in \{0, 1\}^* \mid \text{contains exactly two '1's between every successive '0's} \}$

a) Draw the state diagram of DFA for $L_{21} \cap L_{22} \cap L_{23}$. [Hint: do not use cross product]

b) Draw the state diagram of DFA for L_{24}^* (kleen star)

c) Draw the diagram of DFA for $L_{24}^* \cap L_{25}$ **[small correction done]**

d) Draw the diagram of NFA for $L_{13}, L_{18}, L_{19}, L_{20}$. [NB: Do not draw a DFA. Draw a 'pure' NFA]